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Inventor(s)

SATO; Hiromichi

FUEL CELL COOLING SYSTEM

Abstract

The fuel cell cooling system includes first and second cooling flow paths, a fuel cell, an ion exchanger, a radiator, and an intermediate heat exchanger. When the temperature of the refrigerant in the first cooling flow path is lower than a reference value during power generation of the fuel cell, a first operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant that has passed through the intermediate heat exchanger flows through the fuel cell and the ion exchanger in parallel and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger and the radiator, and when the temperature of the refrigerant in the first cooling flow path is higher than the reference value during the power generation in the fuel cell, a second operation is performed.

Inventors: SATO; Hiromichi (Hadano-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 1000008336368

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032832 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The technology disclosed herein relates to a fuel cell cooling system.

2. Description of Related Art

[0003] A fuel cell system disclosed in Japanese Unexamined Patent Application Publication No. 2002-33108 (JP 2002-33108 A) has a first cooling flow path and a second cooling flow path for circulating a refrigerant. A fuel cell is provided in the first cooling flow path. A radiator is provided in the second cooling flow path. An intermediate heat exchanger is provided across the first cooling flow path and the second cooling flow path. The intermediate heat exchanger exchanges heat between the first cooling flow path and the second cooling flow path.

SUMMARY

[0004] In the fuel cell system, an ion exchanger may be provided to remove ions from the refrigerant flowing through the cooling flow path. In the technology of JP 2002-33108 A, when the ion exchanger is provided in the first cooling flow path, a high-temperature refrigerant flows into the ion exchanger, and the ion exchanger may deteriorate. The present specification proposes a technology for suppressing thermal deterioration of an ion exchanger.

First Aspect

[0005] A fuel cell cooling system disclosed herein includes: [0006] a first cooling flow path through which a refrigerant circulates; [0007] a fuel cell provided in the first cooling flow path; [0008] an ion exchanger provided in the first cooling flow path; [0009] a second cooling flow path through which a refrigerant circulates; [0010] a radiator provided in the second cooling flow path; and [0011] an intermediate heat exchanger configured to perform heat exchange between the first cooling flow path and the second cooling flow path.

[0012] When a temperature of the refrigerant in the first cooling flow path is lower than a reference value during power generation in the fuel cell, a first operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant that has passed through the intermediate heat exchanger flows through the fuel cell and the ion exchanger in parallel and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger and the radiator. When the temperature of the refrigerant in the first cooling flow path is higher than the reference value during the power generation in the fuel cell, a second operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger and the fuel cell while a flow path of the ion exchanger is closed and circulate the refrigerant in the second cooling flow path along the route in which the refrigerant passes through the intermediate heat exchanger and the radiator.

[0013] In the fuel cell cooling system described above, when the temperature of the refrigerant in the first cooling flow path is higher than the reference value during the power generation in the fuel cell, the second operation is performed. In the second operation, the refrigerant circulates in the first cooling flow path along the route in which the refrigerant passes through the intermediate heat exchanger and the fuel cell while the flow path of the ion exchanger is closed. That is, the high-

temperature refrigerant does not flow into the ion exchanger. As a result, thermal deterioration of the ion exchanger can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0015] FIG. 1 is a block diagram of a fuel cell cooling system of Example 1;

[0016] FIG. 2 is a flowchart of an operation selection process of the fuel cell cooling system according to the first embodiment;

[0017] FIG. 3 is a block-diagram of a fuel cell cooling system according to a second embodiment; and

[0018] FIG. 4 is a flowchart of an operation selection process of the fuel cell cooling system according to the second embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0019] Following the above first aspect, additional configurations of the fuel cell cooling system disclosed herein are described below.

Second Aspect

[0020] The fuel cell cooling system according to the first aspect, wherein the temperature of the reference value is 75° C. or higher.

Third Aspect

[0021] In the in the fuel cell cooling system according to the first or second aspect, when the temperature of the refrigerant in the first cooling flow path is higher than the reference value and the conductivity of the refrigerant in the first cooling flow path is higher than a threshold value, the first operation is performed.

Fourth Aspect

[0022] The fuel cell cooling system according to any one of the first to third aspects, wherein the first cooling flow path comprises: [0023] A fuel cell flow path provided with the fuel cell, [0024] An ion exchanger flow path provided with the ion exchanger, [0025] An intermediate heat exchanger flow path provided with the intermediate heat exchanger, wherein [0026] A downstream end of the intermediate heat exchanger flow path is connected to an upstream end of the ion exchanger flow path and an upstream end of the fuel cell flow path, an upstream end of the intermediate heat exchanger flow path is connected to a downstream end of the ion exchanger flow path and a downstream end of the fuel cell flow path, and has a pump provided in the intermediate heat exchanger flow path.

[0027] According to the second aspect, deterioration of the resin of the ion exchanger can be suppressed.

[0028] According to the third aspect, even if the temperature of the refrigerant in the first cooling flow path is high, if the conductivity of the first cooling flow path is high, the refrigerant in the first cooling flow path is caused to flow into the ion exchanger. As a result, it is possible to suppress the increase in the conductivity of the refrigerant while minimizing the thermal deterioration of the ion exchanger.

[0029] According to the fourth aspect, since the refrigerant cooled by the intermediate heat exchanger flows into the ion exchanger, it is possible to more effectively suppress the thermal deterioration of the ion exchanger.

First Embodiment

[0030] The fuel cell cooling system of the first embodiment will be described. The fuel cell cooling

system **100** shown in FIG. **1** is mounted on a device (for example, a fuel cell electric vehicle) using a fuel cell as a power source. The fuel cell cooling system **100** includes a first cooling flow path **12**, a second cooling flow path **22**, an intermediate heat exchanger **30**, a fuel cell **14**, an ion exchanger **32**, and a radiator **24**.

[0031] The intermediate heat exchanger **30** has a first heat exchanger flow path **30a** and a second heat exchanger flow path **30b**. The first heat exchanger flow path **30a** is a part of the first cooling flow path **12**, and the second heat exchanger flow path **30b** is a part of the second cooling flow path **22**. The intermediate heat exchanger **30** causes heat exchange between the first heat exchanger flow path **30a** and the second heat exchanger flow path **30b**. That is, the intermediate heat exchanger **30** causes heat exchange between the first cooling flow path **12** and the second cooling flow path **22**.

[0032] The first cooling flow path **12** includes a fuel cell flow path **12a**, an ion exchanger flow path **12b**, and an intermediate heat exchanger flow path **12c**. A fuel cell **14** is provided in the fuel cell flow path **12a**. An ion exchanger **32** is provided in the ion exchanger flow path **12b**. An intermediate heat exchanger **30** is provided in the intermediate heat exchanger flow path **12c**. Part of the intermediate heat exchanger flow path **12c** is constituted by the first heat exchanger flow path **30a** of the intermediate heat exchanger **30**. The downstream end of the intermediate heat exchanger flow path **12c** is connected to the upstream end of the ion exchanger flow path **12b** and the upstream end of the fuel cell flow path **12a**. Further, the upstream end of the intermediate heat exchanger flow path **12c** is connected to the downstream end of the ion exchanger flow path **12b** and the downstream end of the fuel cell flow path **12a**. A pump **16** is provided in the intermediate heat exchanger flow path **12c**. The pump **16** feeds the refrigerant from the installation position toward the intermediate heat exchanger **30**. When the pump **16** is operated, the refrigerant circulates in the first cooling flow path **12**. The coolant that has passed through the intermediate heat exchanger flow path **12c** flows in parallel to the ion exchanger flow path **12b** and the fuel cell flow path **12a**.

[0033] The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen, and supplies electric power to a motor (not shown). The fuel cell **14** is cooled by the coolant flowing in the fuel cell flow path **12a**.

[0034] The ion exchanger **32** removes ions from the coolant flowing in the ion exchanger flow path **12b**. The ion exchanger **32** lowers the conductivity of the refrigerant by lowering the ion concentration in the refrigerant.

[0035] A valve **38** is provided in the ion exchanger flow path **12b**. The valve **38** is provided between the upstream end of the ion exchanger flow path **12b** and the ion exchanger **32**. The valve **38** opens and closes the flow path of the ion exchanger flow path **12b**. When the valve **38** is open, the coolant that has passed through the intermediate heat exchanger **30** flows in parallel to the ion exchanger flow path **12b** and the fuel cell flow path **12a**. In addition, when the valve **38** is closed, the coolant that has passed through the intermediate **20** heat exchanger **30** does not flow to the ion exchanger flow path **12b** but flows to the fuel cell flow path **12a**.

[0036] The fuel cell cooling system **100** includes temperature sensors **18a**, **18b**. The temperature sensor **18a** is provided upstream of the fuel cell flow path **12a**. The temperature sensor **18a** detects the temperature of the coolant flowing into the fuel cell **14** (hereinafter referred to as FC inlet temperature). The temperature sensor **18b** is provided downstream of the fuel cell flow path **12a**. The temperature sensor **18b** detects the temperature of the coolant discharged from the fuel cell **14**.

[0037] The second cooling flow path **22** is an annular flow path. The radiator **24** is provided in the second cooling flow path **22**. The radiator **24** cools the refrigerant flowing through the second cooling flow path **22** by heat exchange with the outside air. In addition, an intermediate heat exchanger **30** is provided in the second cooling flow path **22**. A part of the second cooling flow path **22** is constituted by the second heat exchanger flow path **30b** of the intermediate heat exchanger **30**.

[0038] The fuel cell cooling system **100** includes a pump **26**, a temperature sensor **28a**, and a temperature sensor **28b**.

[0039] The pump **26** is provided in the second cooling flow path **22**. The pump **26** delivers the refrigerant from the installation position toward the intermediate heat exchanger **30**. When the pump **26** is operated, the refrigerant circulates in the second cooling flow path **22**.

[0040] The temperature sensor **28a** is provided downstream of the radiator **24** in the second cooling flow path **22**. The temperature sensor **28a** detects the temperature of the coolant emitted from the radiator **24**. The temperature sensor **28b** is provided in the second cooling flow path **22** upstream of the radiator **24**. The temperature sensor **28b** detects the temperature of the coolant flowing into the radiator **24**.

[0041] The fuel cell cooling system **100** includes a control device **40**. The control device **40** controls the pump **16**, the pump **26**, and the valve **38** based on the temperature of the refrigerant in the first cooling flow path **12**.

[0042] When the fuel cell **14** generates electricity, the fuel cell **14** generates heat. The control device **40** activates the pumps **16**, **26** during power generation of the fuel cell **14**. When the pump **16** is operated, the refrigerant circulates in the first cooling flow path **12**. When the pump **26** is operated, the refrigerant circulates in the second cooling flow path **22**. The fuel cell **14** is cooled by heat exchange between the refrigerant in the first cooling flow path **12** and the fuel cell **14**. The heat generated by the power generation of the fuel cell **14** is transferred to the intermediate heat exchanger **30** (that is, the first heat exchanger flow path **30a**) by the coolant in the first cooling flow path **12**. The intermediate heat exchanger **30** transfers the heat generated by the power generation of the fuel cell **14** from the first heat exchanger flow path **30a** to the second heat exchanger flow path **30b**. Heat is transferred from the intermediate heat exchanger **30** to the radiator **24** by the refrigerant in the second cooling flow path **22**. The refrigerant in the second cooling flow path **22** is cooled by the radiator **24** performing heat exchange with the outside air.

[0043] The fuel cell cooling system **100** controls the valve **38** according to the flowchart shown in FIG. **2** during power generation of the fuel cell **14**.

[0044] In **S2**, the control device **40** determines whether FC inlet temperature is equal to or lower than the determination temperature **T1**. The determination temperature **T1** is set to a temperature (for example, 75° C. or higher) at which the performance of the ion exchanger **32** can be ensured. As the determination temperature **T1**, for example, the degradation-onset temperature of the resin in the ion exchanger **32** can be adopted.

[0045] If FC inlet temperature is less than or equal to the determination temperature **T1** (**S2** is YES), the control device **40** brings the valve **38** into an open state (**S4**). When the valve **38** is opened, the refrigerant that has passed through the intermediate heat exchanger **30** flows in parallel to the ion exchanger **32** and the fuel cell **14** in the first cooling flow path **12**. In this case, since the temperature of the refrigerant in the first cooling flow path **12** is low, the low-temperature refrigerant flows into the ion exchanger **32**. Therefore, in the ion exchanger **32**, deterioration due to heat of the flowing refrigerant is less likely to occur. In particular, since the low-temperature refrigerant immediately after passing through the intermediate heat exchanger **30** flows into the ion exchanger **32**, the thermal deterioration of the ion exchanger **32** can be effectively suppressed. In addition, when the determination temperature **T1** is set to the deterioration starting temperature of the resin, the deterioration of the resin in the ion exchanger **32** can be effectively suppressed. When the refrigerant flows into the ion exchanger **32**, the ion exchanger **32** removes ions from the refrigerant. As a result, the ion concentration in the refrigerant decreases, and the conductivity of the refrigerant decreases. Further, the fuel cell **14** is efficiently cooled by the inflowing refrigerant.

[0046] The control device **40** repeatedly executes **S6** from **S2** while FC inlet temperature is equal to or lower than the determination temperature **T1**. Therefore, while FC inlet temperature is equal to or lower than the determination temperature **T1**, the control device **40** opens the valve **38** until the power of the vehicle is turned OFF.

[0047] If FC inlet temperature exceeds the determination temperature T1 (NO at S2), the control device 40 brings the valve 38 into a closed state (S8). When the valve 38 is closed, the refrigerant that has passed through the intermediate heat exchanger 30 does not flow to the ion exchanger 32 in the first cooling flow path 12. Therefore, in the first cooling flow path 12, the refrigerant circulates through the intermediate heat exchanger 30 and the fuel cell 14. In this way, when the temperature of the refrigerant in the first cooling flow path 12 is high, the valve 38 is closed to prevent the high-temperature refrigerant from flowing into the ion exchanger 32. Therefore, in the ion exchanger 32, deterioration due to heat of the refrigerant is less likely to occur. Further, the fuel cell 14 is cooled by a refrigerant.

[0048] The control device 40 repeatedly executes S2, S8, S10 while FC inlet temperature exceeds the determination temperature T1. Thus, while FC inlet temperature exceeds the determination temperature T1, the control device 40 closes the valve 38 until the vehicle is powered OFF.

[0049] As described above, in the first embodiment, the control device 40 closes the valve 38 when FC inlet temperature is high, thereby reducing the high-temperature coolant from flowing into the ion exchanger 32. Therefore, deterioration of the ion exchanger 32 can be suppressed.

Second Embodiment

[0050] The fuel cell cooling system of the second embodiment will be described. As shown in FIG. 3, the fuel cell cooling system 200 of the second embodiment has a configuration in which a conductivity meter 50 is added to the fuel cell cooling system 100 of the first embodiment. The conductivity meter 50 measures the conductivity C2 of the coolant flowing through the first cooling flow path 12.

[0051] The fuel cell cooling system 200 controls the valve 38 according to the flowchart shown in FIG. 4 during power generation of the fuel cell 14. S2, S4, S6 and S8 shown in FIG. 4 are the same as S2, S4, S6 and S8 shown in FIG. 2.

[0052] In the second embodiment, after S8 is performed, the control device 40 determines whether the conductivity C2 exceeds the conductivity upper limit C1 in S20.

[0053] If the conductivity C2 exceeds the conductivity upper limit C1 (S20 and YES), the control device 40 brings the valve 38 into an open state (S22). When the valve 38 is opened, the refrigerant that has passed through the intermediate heat exchanger 30 flows in parallel to the ion exchanger 32 and the fuel cell 14 in the first cooling flow path 12. When the refrigerant flows into the ion exchanger 32, the ion exchanger 32 removes ions from the refrigerant. Accordingly, the ion concentration in the refrigerant can be reduced. As a consequence, the conductivity C2 of the coolant flowing in the first cooling flow path 12 decreases.

[0054] The control device 40 repeatedly executes S24 from S20 while the conductivity C2 exceeds the conductivity upper limit C1. Thus, while the conductivity C2 exceeds the conductivity upper limit C1, the control device 40 opens the valve 38 until the vehicle is powered OFF. This lowers the conductivity C2.

[0055] If the conductivity C2 is less than the conductivity upper limit C1 (NO by S20), the control device 40 keeps the valve 38 closed. Therefore, in the first cooling flow path 12, the refrigerant that has passed through the intermediate heat exchanger 30 does not flow to the ion exchanger 32. Therefore, in the first cooling flow path 12, the refrigerant circulates through the intermediate heat exchanger 30 and the fuel cell 14. As described above, when the temperature of the refrigerant in the first cooling flow path 12 is high and the conductivity C2 is less than the conductivity upper limit C1, the control device 40 reduces the high-temperature refrigerant from flowing into the ion exchanger 32 by closing the valve 38. Therefore, in the ion exchanger 32, deterioration due to heat of the refrigerant is less likely to occur. The control device 40 repeatedly executes S2, S8, S20 and S26 while the conductivity C2 is less than the conductivity upper limit C1. Therefore, while the conductivity C2 is less than the conductivity upper limit C1, the control device 40 closes the valve 38 until the power of the vehicle is turned OFF.

[0056] As described above, in the second embodiment, the control device 40 opens the valve 38

when the conductivity C2 of the refrigerant in the first cooling flow path 12 is higher even when the temperature of the refrigerant in the first cooling flow path 12 is high. Thus, the conductivity C2 of the coolant flowing in the first cooling flow path 12 can be suppressed from becoming excessively high.

[0057] In first and second embodiments, the flow path changing device for changing the flow path of the refrigerant in the first cooling flow path 12 by the valve 38 is configured, the flow path changing device may be configured by other devices (for example, a three-way valve or the like). Further, in the first embodiment, the valve 38 is a solenoid valve, but a flow path changing device may be configured by a device that does not use electricity. For example, a flow path changing device may be configured by a device that switches a flow path by thermal expansion of a material, such as a thermostat.

[0058] The process of opening the valve 38 of the first and second embodiments is an example of the first operation. The process of closing the valve 38 of the first and second embodiments is an example of the second operation.

[0059] The determination temperature T1 of the first and second embodiments are examples of reference values. The conductivity upper limit C1 of the second embodiment is an example of thresholds.

[0060] While the embodiments have been described in detail above, these are merely illustrative and do not limit the scope of the claims. The technology described in the claims includes various modifications and alterations of the specific examples described above. The technical elements described in this specification or in the drawings may be used alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. Further, the technology illustrated in the present specification or the drawings achieves a plurality of objects at the same time, and has technical usefulness by achieving one of the objects.

Claims

1. A fuel cell cooling system comprising: a first cooling flow path through which a refrigerant circulates; a fuel cell provided in the first cooling flow path; an ion exchanger provided in the first cooling flow path; a second cooling flow path through which a refrigerant circulates; a radiator provided in the second cooling flow path; and an intermediate heat exchanger configured to perform heat exchange between the first cooling flow path and the second cooling flow path, wherein when a temperature of the refrigerant in the first cooling flow path is lower than a reference value during power generation in the fuel cell, a first operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant that has passed through the intermediate heat exchanger flows through the fuel cell and the ion exchanger in parallel and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger and the radiator, and when the temperature of the refrigerant in the first cooling flow path is higher than the reference value during the power generation in the fuel cell, a second operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger and the fuel cell while a flow path of the ion exchanger is closed and circulate the refrigerant in the second cooling flow path along the route in which the refrigerant passes through the intermediate heat exchanger and the radiator.
2. The fuel cell cooling system according to claim 1, wherein the reference value is 75° C. or higher.
3. The fuel cell cooling system according to claim 1, wherein the first operation is performed when the temperature of the refrigerant in the first cooling flow path is higher than the reference value and conductivity of the refrigerant in the first cooling flow path is higher than a threshold value.
4. The fuel cell cooling system according to claim 1, wherein: the first cooling flow path includes:

a fuel cell flow path provided with the fuel cell; an ion exchanger flow path provided with the ion exchanger; and an intermediate heat exchanger flow path provided with the intermediate heat exchanger; a downstream end of the intermediate heat exchanger flow path is connected to an upstream end of the ion exchanger flow path and an upstream end of the fuel cell flow path; an upstream end of the intermediate heat exchanger flow path is connected to a downstream end of the ion exchanger flow path and a downstream end of the fuel cell flow path; and a pump is provided in the intermediate heat exchanger flow path.
